NIST binary curves: skipped candidates (2026-10-20)

Status. Measured census of how the FIPS 186-2 binary curves were chosen. No bearing on ECDLP hardness. Study directory: `research/nist_binary_skipped_20261020/` (`census.sh`, about 30 s; `cargo test --release --lib nist_seed_provenance`, 8 tests).

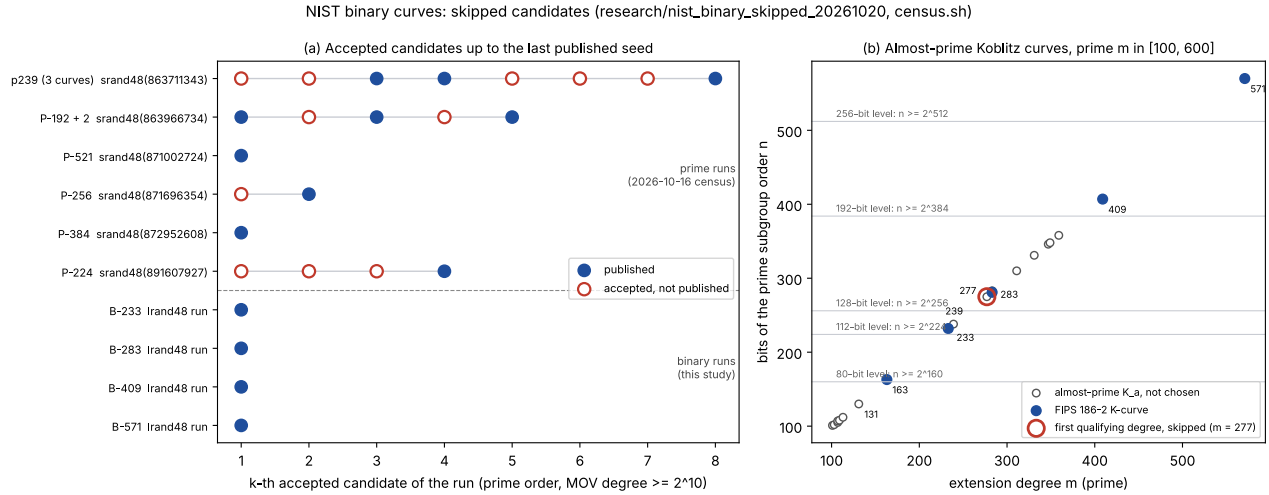


Figure 1: (a) Accepted candidates of each seed run up to its last published seed: prime runs from the 2026-10-16 census, binary runs from this study (`census/summary.json`). (b) Almost-prime Koblitz curves for prime $m \in [100, 600]$ with the FIPS thresholds $n \geq 2^{2s}$ (`census/koblitz_sweep.txt`).

Method

- Candidates (Rust, `examples/nist_binary_seed_candidates.rs`): candidate i is five `lrand48()` words after `srand48(run)`; its X9.62 A.3.3.2 value is $b_0 \parallel \text{SHA-1}(\text{seed} + 1) \parallel \ldots$, with b_0 the rightmost $m - 160s$ bits of SHA-1(seed).
- Basis (GP, `census.gp`): the value is b in the Gaussian normal basis. The root β of the Gauss period's minimal polynomial in $\text{GF}(2)[z]/(f)$ that maps each published seed's value to the published polynomial-basis b was unique for all five FIPS curves (B-163 included), all with the same bit order. Rotating the basis is a Frobenius, so the orders do not depend on the root.
- Point counting: PARI/GP 2.15.4 `ellcard`. Acceptance: $\#E = 2n$, n prime, MOV degree $\geq 2^{10}$. Checks on every row: $\#E \equiv 2 \pmod 4$, Hasse bound, $[\#E]P = O$ at three random points.
- Koblitz sweep (Rust, `examples/koblitz_degree_sweep.rs`): Lucas recurrence, Miller–Rabin with 20 bases.

Results (measured)

run	curve	candidates	checks pass	accepted	published
446413329	B-233	155	155	155	155
446413432	B-283	32	32	32	32
446413509	B-409	60	60	60	60
446413564	B-571	73	73	73	73

In every binary run the published seed is the **first** accepted candidate; no B-curve candidate was skipped. Positive control: at the published index the pipeline reproduces the published polynomial-basis b and $\#E = 2n$ for all four.

level s	need bits(n)	smallest qualifying m	FIPS m
80	160	163	163
112	224	233	233
128	**256**	**277**	**283**
192	384	409	409
256	512	571	571

The one unexplained Koblitz skip is $y^2 + xy = x^3 + 1$ over $\mathrm{GF}(2^{277})$ ($a = 0$): $\#E = 4n$, n a 275-bit prime, least GNB type 4 (K-283: type 6), no trinomial (as for 163, 283, 571), $a = 0$ like the four large K-curves. ICV1: `icv1-f2m277-t412886371405096939848969796546990392459541-c0810384` (least sparse irreducible modulus; no published polynomial). At $m = 283$, K_1 also qualifies with a larger n (283 vs 281 bits); FIPS took K_0.

Derived (not measured)

Prime runs: 21 accepted, 10 published. With each accepted candidate published independently with probability 10/21, four runs all publishing their first has probability $(10/21)^4 \approx 0.05$ — a contrast, not a test. Expected first acceptance index $\approx (m-1)\ln 2/2$: 80, 98, 141, 197 for m = 233, 283, 409, 571; observed 155, 32, 60, 73.

Open

The B-163 seed; binary base points; why $m = 283$ over $m = 277$; the prime-curve skip filter.